BLOCKCHAIN-BASED PROPERTY OWNERSHIP AND LAND RECORD MANAGEMENT SYSTEM

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

GUJJULA MANASA

23501A0548

Under the guidance of

Mr. A. Prashant, Asst. Prof.



Department of Computer Science and Engineering

Prasad V Potluri Siddhartha Institute of Technology

(Permanently affiliated to JNTU-Kakinada, Approved by AICTE)

(An NBA & NAAC accredited and ISO 9001:2015 certified institute)

Kanuru, Vijayawada-520 007

2026-27

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CERTIFICATE

This is to certify that the Use Case report entitled **"Blockchain-Based Property Ownership and Land Record Management System"** that is being submitted by **Gujjula Manasa (23501A0548)**, as part of Assignment-2 for the **Blockchain Technology (23CS4701B)** course in 4**-1** during the academic year **2026-27**.

Course Coordinator
Mr. A. Prashant
Assistant Professor,
Department of CSE,
PVPSIT, Vijayawada

Head of the Department
Dr. A. Jayalakshmi,
Professor and Head,
Department of CSE,
PVPSIT, Vijayawada

MARKS
Assignment 2 : /5

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain technology is a decentralized and distributed ledger system that records transactions across a network of independent computers, ensuring transparency, security, and immutability. Unlike a conventional database that is controlled by a single organization, a blockchain operates on a peer-to-peer network in which every participant holds a synchronized copy of the ledger. Each block in the chain stores a cryptographic hash of the block before it, so altering any earlier record would break every block that follows and be detected immediately. Although blockchain was first introduced for cryptocurrency transactions, it is now being applied in finance, supply chain management, healthcare, education, and public administration, including land and property record management. [3]

1.2. Relevance of Blockchain in Land Record Management

Land is one of the most valuable assets a person can own, and the record of who owns it must therefore be accurate, secure, and trusted by everyone. In most places today, land records are maintained by revenue departments and sub-registrar offices in the form of paper registers, registered sale deeds, and centralized digital databases. These systems are prone to problems such as lost or damaged documents, forged deeds, delays in updating ownership, and the manipulation of entries by insiders or intermediaries. The Andhra Pradesh government's own IT adviser has described the existing system as corrupt, estimated that around $700 million is paid in bribes at land registrars across India, and pointed out that land and property matters make up about two-thirds of all civil cases in the country. [1]

A blockchain-based land record management system addresses these issues by digitizing land titles and storing the ownership record on a decentralized ledger that no single official can quietly alter. By using smart contracts, the steps of registering a parcel, verifying its owner, and transferring it to a buyer can be enforced by code instead of being left entirely to manual processing. Documents such as the sale deed, survey sketch, and encumbrance details can be stored off-chain, while their cryptographic hash is recorded on the blockchain so that any later change to a document is immediately visible. [3] [5]

Governments have already begun to explore this idea. The Government of Andhra Pradesh partnered with the Swedish company ChromaWay in 2017 on a land registry pilot that uses a distributed ledger at the back end and a simple web application at the front end, as part of the Fintech Valley Vizag initiative. [2] Telangana started pilots for land registration with the Centre for Development of Advanced Computing (C-DAC) [9] and began evaluating blockchain to secure its digitized revenue and land records [6], and NITI Aayog identified land records as one of the priority areas for blockchain in governance. [7] These efforts show that blockchain-based land administration is a realistic and relevant direction for the country.

Furthermore, because the ledger is shared among all authorized parties, a buyer, a bank, a court, and a revenue officer can each verify the same ownership record without depending on a single office or on paper certificates that may have been tampered with. This is particularly valuable in India, where the registration of a sale and the record of rights are often maintained in separate systems, which makes it difficult for a buyer to trace the full history of a property. [7] A blockchain-based land record system links these steps into one verifiable chain of ownership,

reduces the opportunity for fraud, shortens the time taken for registration and mutation, and builds public trust in the land administration process. As governments continue to digitize public services, blockchain offers a scalable and future-proof foundation for managing property ownership securely and transparently.

2. BACKGROUND

Traditional land record management systems face numerous challenges that lead to fraud, long disputes, and a loss of public confidence. These challenges arise from the reliance on paper-based documents, centralized databases, and manual processing carried out by several disconnected departments. Below are some key issues with the existing land record systems:

2.1. Paper-Based Records and Physical Deeds

Many land records still depend on paper registers, handwritten entries, and physical sale deeds. These documents are easily lost, damaged by fire, floods, or poor storage, and often become illegible with time. When the original record is missing, owners find it extremely difficult to prove their title, which leads to long delays in sale, inheritance, and bank loan processes.

2.2. Fraudulent Registrations and Forged Documents

Because it is hard to verify the authenticity of a paper document, forged sale deeds, fake ownership certificates, and altered survey details are common. Fraudsters may impersonate real owners or create false documents to sell land that does not belong to them, causing heavy financial losses to innocent buyers and long legal battles for the real owners. [3]

2.3. Multiple Sale of the Same Property

When there is no single, tamper-proof and instantly updated record, the same parcel of land can be sold to more than one buyer. Each buyer may hold a seemingly valid document, and the dispute then has to be settled in court. Preventing the trading of one piece of land to several buyers is one of the main problems that blockchain-based land registries aim to solve. [3]

2.4. Corruption and Dependence on Intermediaries

Land registration usually involves several officials and agents who control access to the records. This gives room for bribes and unofficial payments to speed up or manipulate entries. It has been estimated that a very large amount is paid in bribes at land registrars across India every year, and that the current system is widely seen as corrupt. [1]

2.5. Delays in Registration and Mutation

After a sale is registered, the revenue record must be updated in the name of the new owner, a process known as mutation. This often requires several visits, repeated verification of documents, and approval at multiple levels, and can take weeks or months. During this period the official record still shows the old owner, which creates confusion and openings for misuse.

2.6. Disconnected Registration and Ownership Records

In many states the registration of documents and the maintenance of the record of rights are handled by different departments using different systems. NITI Aayog has observed that these two are not properly linked, which makes due diligence in a property transaction difficult and leaves buyers without easy access to the historical ownership details of a parcel. [7]

2.7. Lack of Transparency and Traceability

Centralized land databases are usually accessible only to the department that maintains them. Citizens, banks, and courts cannot independently verify who changed a record, when it was changed, or on what basis. This lack of an auditable trail makes it hard to detect unauthorized modifications and creates mistrust between landowners and the administration.

2.8. Land Disputes and Litigation Burden

Unclear titles, overlapping claims, and boundary disagreements result in a very large number of land-related cases. Land and property matters are estimated to account for around two-thirds of all civil cases in India, which places a heavy burden on the judiciary and keeps both parties tied up in litigation for years. [1]

2.9. Security Risks and Tampering of Centralized Databases

Even when records are digitized, they are often kept in a central database under the control of a few administrators. Anyone with access, from a data entry operator to an external attacker, can potentially change an entry. Reports on the Andhra Pradesh initiative noted that the records could still be tampered with before they are entered on a blockchain, and that landowners had no way to monitor changes made to their records on a private chain. [8]

2.10. Limited Citizen Access to Land Information

Ordinary citizens often have to visit government offices or approach intermediaries to obtain copies of their records, encumbrance details, or the status of a pending application. The lack of simple, real-time, and user-friendly access increases cost, delays, and dependence on middlemen, especially for farmers and rural landowners.

Due to these challenges, there is a strong need for a secure, transparent, and automated land record management system. Blockchain technology offers a promising solution by providing tamper-proof records, verifiable ownership history, and automated rules for registration and transfer without depending on a single point of trust.

3. BLOCKCHAIN BASICS

Blockchain is a distributed ledger technology that enables secure, transparent, and tamper-proof recording of transactions without relying on a trusted central authority. Instead of storing data on one server, a blockchain replicates it across many nodes that follow common rules to agree on the state of the ledger. This makes it well suited to applications such as land registries, where the main requirement is a record that many parties can trust. Below are the key concepts that define blockchain technology:

3.1. Decentralization

Traditional land databases are controlled by a single department, which creates a single point of failure and a single point of manipulation. A blockchain, on the other hand, runs on a decentralized network where multiple independent nodes validate and store the data. No single participant has full control, and compromising one node does not affect the integrity of the entire network. For a land registry, the nodes can be operated by different stakeholders such as the revenue department, the registration department, and the National Informatics Centre, so that no single office can change ownership entries alone. [3]

3.2. Immutability

Once a transaction is recorded on a blockchain and confirmed by the network, it cannot be altered or deleted. This is achieved through cryptographic hashing, where each block contains the hash of the previous block, forming a chain. Changing one record would require changing every later block on a majority of nodes, which is practically impossible. This property is extremely useful for land records, because the history of every ownership change is preserved permanently, and any correction has to be made as a new, visible transaction instead of silently overwriting the old one.

3.3. Transparency

In a blockchain network, every authorized participant holds an identical copy of the ledger, and every transaction can be audited. This transparency makes it possible for buyers, banks, and courts to verify the ownership and transfer history of a property independently, without depending on the department that maintains the record. Any attempt to manipulate a record would be quickly detected by the network because the copies held by other nodes would no longer match. [4]

3.4. Smart Contracts

Smart contracts are self-executing programs stored on the blockchain that run automatically when predefined conditions are met. In a land registry, smart contracts can enforce rules such as registering a survey number only once, allowing only the current owner to start a sale, holding the buyer’s payment in escrow, and releasing it to the seller only after the registrar approves the transfer. Because these rules are executed by code on the network, they are applied in the same way every time, which reduces human error, delays, and opportunities for misconduct. [3] [5]

3.5. Consensus Mechanisms

Consensus mechanisms are the rules by which the nodes of a blockchain agree on which transactions are valid and in which order they are added. In Proof of Work (PoW), used by Bitcoin, miners compete to solve computational puzzles. In Proof of Stake (PoS), which Ethereum has used since 2022, validators are chosen according to the amount of cryptocurrency they have locked up as a stake. Permissioned networks often use lighter mechanisms such as Proof of Authority (PoA) or practical Byzantine fault tolerant protocols, where a limited set of known and approved validators confirm the transactions. For government land records, a permissioned or hybrid design with known validators is often preferred because it offers better speed and accountability.

3.6. Cryptographic Security

Blockchain security is based on public-key cryptography and hash functions. Every user has a private key that is used to digitally sign transactions and a public key that is used to verify the signature, so only the genuine owner of an account can initiate an action from it. Hash functions convert any document into a fixed-length digital fingerprint, so even a one-character change in a sale deed produces a completely different hash. Storing the hash of each land document on the blockchain therefore allows anyone to confirm later that the document has not been modified. [4]

3.7. Public and Permissioned Blockchains

Blockchain networks can be public, where anyone can join and read the ledger, or permissioned, where only approved participants can write to or read from it. Public networks such as Ethereum offer maximum transparency and resistance to censorship, but they expose data to everyone and charge transaction fees. Permissioned networks give government departments control over who can participate and help to protect personal information, but they require the participants to trust the operators of the network. Many land registry pilots, including the one in Andhra Pradesh, have therefore used private or permissioned designs. [8] The prototype in this report is developed on Ethereum tools so that its working can be demonstrated and tested easily, and it can be moved to a permissioned Ethereum-compatible network for real deployment.

By combining these features—decentralization, immutability, transparency, smart contracts, and cryptographic security—blockchain provides a robust foundation for land record management. It ensures that ownership records are secure, verifiable, and updated through transparent and automated processes, which increases trust among owners, buyers, and the authorities.

4. USE CASE OVERVIEW

The use case for a Blockchain-Based Property Ownership and Land Record Management System aims to transform traditional land administration by using blockchain technology. The system provides a secure, transparent, and fraud-resistant way to register land, verify ownership, and transfer property from one owner to another.

4.1. Objectives

The primary objective of this system is to replace paper-based and loosely connected land records with a single digital platform for registration and verification of land ownership. Each parcel of land is represented by a unique record on the blockchain that holds its survey number, location, area, current owner, and the hash of its supporting documents. This allows ownership to be verified at any time without visiting an office or depending on physical papers. [3]

A key challenge in land administration is forgery and the sale of the same land more than once. The smart contract allows a survey number to be registered only a single time and allows only the current owner to initiate a sale. Since the ledger is immutable, a registered record cannot be altered or deleted, which significantly reduces fraudulent claims and double selling.

Another important goal is to make property transfer simple and secure. When a sale is agreed, the owner initiates the transfer on the platform, the buyer deposits the agreed amount into the contract, and the sub-registrar verifies the documents and approves the transfer. On approval, ownership changes and the money is released to the seller in the same transaction, so neither party has to trust the other, and the revenue record is updated immediately instead of after weeks of mutation.

The system also improves transparency and traceability. Owners, buyers, banks, and officials can access the same up-to-date information and the complete ownership history of any parcel, which removes the need to depend on intermediaries and reduces disputes over who owns what. [7]

By automating verification and reducing manual handling of files, the system can lower the operational cost of land administration and the opportunities for corruption. Security is another key focus, because centralized land databases are vulnerable to unauthorized changes. Storing ownership records on a decentralized network, with every change tied to a digitally signed transaction, makes tampering far more difficult and every change traceable to the account that made it.

4.2. Scope of the System

The blockchain-based land record system focuses on registering, managing, verifying, and transferring land ownership. The system includes:

1. **Sub-Registrar / Revenue Officials (Registrars):** Authorized government officers who register land parcels after verifying the documents and who approve or cancel ownership transfers.
2. **Land Owners:** Citizens who view the land registered in their name, check its details, and initiate a transfer when they sell the property.

3. **Buyers:** Prospective purchasers who verify the ownership of a parcel before buying and who deposit the payment for an approved transaction.
4. **Blockchain Network:** A decentralized ledger that stores tamper-proof ownership records and executes the smart contract rules.

4.3. System Architecture

The architecture of the Blockchain-Based Property Ownership and Land Record Management System is illustrated in Figure 4.1 and consists of the following components:

4.3.1. User Layer

This layer consists of the primary users who interact with the system. (Figure 4.1 - User Layer).

- **Land Owners:**
  - View the land parcels registered in their name.
  - Initiate the transfer of a parcel when it is sold.
- **Buyers:**
  - Verify the ownership and details of a parcel before purchase.
  - Deposit the agreed payment for a transfer.
- **Sub-Registrar / Revenue Officials:**
  - Register new land parcels after verifying the supporting documents.
  - Approve valid transfers or cancel invalid ones.

4.3.2. Web Application Layer

This layer acts as the interface between users and the blockchain. (Figure 4.1 - Web Application Layer).

- **User Interface:**
  - Allows registrars to register land and process transfer requests.
  - Enables owners and buyers to view, verify, and transfer land records.
- **Land Management System:**
  - Handles user authentication and wallet connection.
  - Facilitates communication between users, the blockchain, and the off-chain document store.

4.3.3. Blockchain Layer

This layer ensures security, transparency, and immutability of land transactions. (Figure 4.1 - Blockchain Layer).

- **Smart Contract Layer:**
  - **LandRegistry Contract:**
    - Stores land details such as survey number, location, area, owner, and document hash.
    - Manages registration, verification, and transfer of ownership.
  - **Smart Contract Execution:**
    - When a registrar registers a parcel, the contract records it on the blockchain.

- When a transfer is approved, ownership moves to the buyer and the payment is released to the seller.

- **Blockchain Network:**
  - **Distributed Ledger:** Maintains a decentralized, tamper-proof record of all land parcels and their owners.
  - **Immutable Transactions:** Ensures ownership data cannot be modified, preventing fraud.
  - **Off-chain Storage (IPFS):** Holds the scanned documents, while only their hash is stored on the blockchain to keep the ledger light. [5]

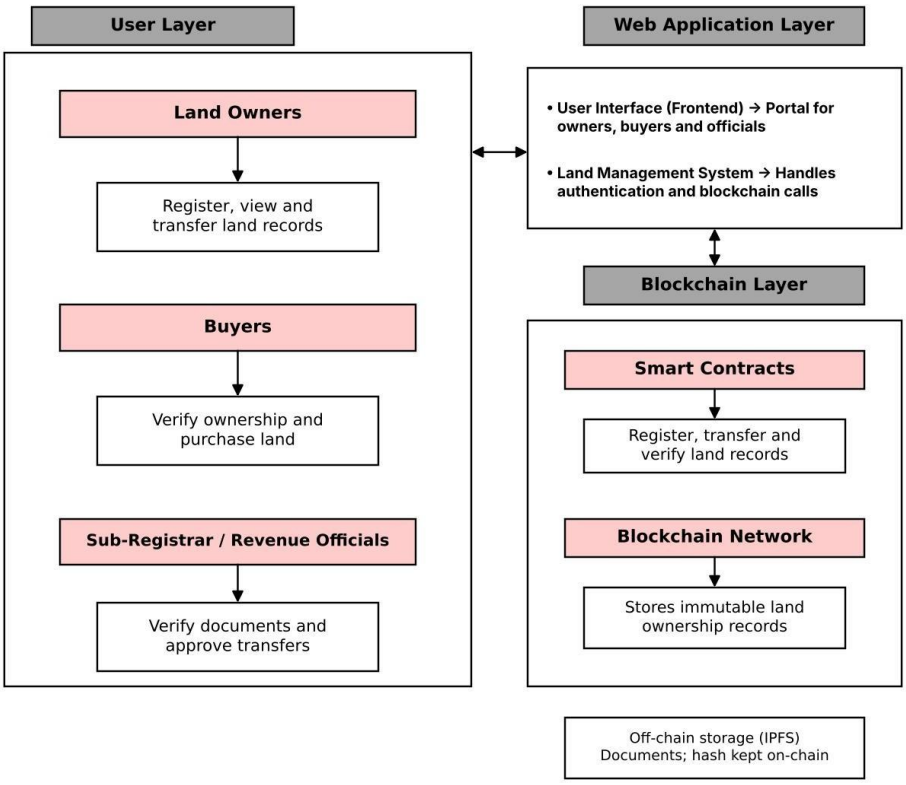


Figure 4.1: Architecture of Blockchain based Land Record Management System

5. IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

To implement the blockchain-based land record system, the development environment is set up using Ethereum as the blockchain platform. Essential tools include Truffle, which serves as a development framework for Ethereum smart contracts, and Ganache, a personal Ethereum blockchain for local testing. MetaMask is integrated into the system to manage blockchain accounts and sign transactions. The smart contract is written in Solidity (v0.8.19), whose built-in overflow checks and improved error handling add to the safety of the code. [10] Once the environment is ready, the smart contract for land management can be written.

5.2. Defining Smart Contracts

5.2.1. Writing the Smart Contract for Land Records

The LandRegistry contract is responsible for registering land, verifying ownership, and transferring parcels. It defines a Land struct that holds the essential details of a parcel: its ID, survey number, current owner, area, registration timestamp, a flag showing whether a transfer is in progress, the location, and the hash of the supporting documents. A second struct, TransferRequest, stores the buyer, the agreed price, and the payment status of a pending sale. The contract also uses mappings to track authorized registrars, registered survey numbers, the land held by each owner, pending transfers, and the full ownership history of every parcel. By structuring the contract this way, every parcel stays linked to its rightful owner, and ownership remains transparent and traceable.

```
struct Land {
    uint256 id;
    uint256 surveyNumber;
    address owner;
    uint256  area;
    uint256 registeredAt;
    bool underTransfer;
    string location;
    string documentHash;
}

struct TransferRequest
    { address buyer;
    uint256 price;
    bool paid;
    bool active;
}

mapping(address => bool) internal registrars;
mapping(uint256 => Land) internal lands;
mapping(uint256 => bool) internal surveyRegistered;
mapping(address => uint256[]) internal ownerToLands;
mapping(uint256 => TransferRequest) internal requests;
mapping(uint256 => address[]) internal ownershipHistory;
```

5.2.2. Registering Authorized Registrars

To prevent unauthorized persons from creating or changing land records, only addresses approved as registrars can register land or approve transfers. The account that deploys the contract becomes the administrator and is automatically a registrar. The administrator can add more registrars, such as the sub-registrars of different offices, through the addRegistrar function.

The onlyRegistrar modifier is then applied to every sensitive function, so that a request from any other account is rejected.

```
modifier onlyRegistrar() {
    require(registrars[msg.sender], "Not an authorised registrar");
    _;
}

constructor() {
    admin = msg.sender;
    registrars[msg.sender] = true;
}

function addRegistrar(address registrar) external
    { require(msg.sender == admin, "Only admin");
    require(registrar != address(0), "Invalid address");
    registrars[registrar] = true;
}
```

5.2.3. Registering Land Records

After verifying the owner's identity and documents off-chain, a registrar registers the parcel by calling registerLand with the owner's blockchain address, the survey number, location, area, and the hash of the uploaded documents. The function rejects a survey number that has already been registered, which prevents the same parcel from being recorded twice. It then assigns a unique land ID, stores the record, adds the parcel to the owner's list, starts the ownership history, and emits an event so that the front end can display the new record.

```
function
    registerLand( address
    owner, uint256
    surveyNumber,
    string memory location,
    uint256 area,
    string memory documentHash
) external onlyRegistrar {
    require(owner != address(0), "Invalid owner");
    require(area > 0, "Invalid area");
    require(!surveyRegistered[surveyNumber], "Survey already registered");
    uint256 landId = nextLandId++;
    surveyRegistered[surveyNumber] = true;
    Land storage l = lands[landId];
    l.id = landId;
    l.surveyNumber = surveyNumber;
    l.owner = owner;
    l.area = area;
    l.registeredAt = block.timestamp;
    l.location = location;
    l.documentHash = documentHash;
    ownerToLands[owner].push(landId);
    ownershipHistory[landId].push(owner);
    emit LandRegistered(landId, owner, surveyNumber);
}
```

5.2.4. Viewing and Verifying Land Records

Owners need to see the land registered in their name, and buyers need to check a parcel before purchase. The viewMyLands function loops through the parcels linked to the caller's address and returns their details. The getLand and verifyOwnership functions allow anyone to look up a parcel and confirm whether a given address is its current owner, while

getOwnershipHistory returns every owner the parcel has had since registration. These are read-only functions, so they cost nothing to call and can be used freely by buyers, banks, and courts.

```
function viewMyLands() external view returns (Land[] memory)
    { uint256[] memory ids = ownerToLands[msg.sender];
    Land[] memory owned = new Land[](ids.length);
    for (uint256 i = 0; i < ids.length; i++) {
        owned[i] = lands[ids[i]];
    }
    return owned;
}

function getLand(uint256 landId) external view returns (Land memory)
    { require(lands[landId].id != 0, "Land not found");
    return lands[landId];
}

function verifyOwnership(uint256 landId, address claimedOwner)
    external view returns (bool)
{
    return lands[landId].id != 0 && lands[landId].owner == claimedOwner;
}

function getOwnershipHistory(uint256 landId)
    external view returns (address[] memory)
{
    return ownershipHistory[landId];
}
```

5.2.5. Initiating a Transfer and Depositing Payment

When a sale is agreed, the current owner calls initiateTransfer with the land ID, the buyer's address, and the agreed price. The function checks that the caller is the rightful owner, that no other transfer is in progress for the parcel, and that the buyer address and price are valid, and then marks the parcel as under transfer. The buyer then calls depositPayment and sends exactly the agreed amount to the contract, where it is held in escrow until the registrar takes a decision. This ensures that the seller is paid only if the transfer is approved and the buyer gets the money back if it is not. [5]

```
function
    initiateTransfer( uin
    t256 landId, address
    buyer, uint256 price
) external {
    Land storage land = lands[landId];
    require(land.owner == msg.sender, "Not the owner");
    require(!land.underTransfer, "Transfer already in progress");
    require(buyer != address(0) && buyer != msg.sender, "Invalid buyer");
    require(price > 0, "Invalid price");
    land.underTransfer = true;
    requests[landId] = TransferRequest(buyer, price, false, true);
    emit TransferInitiated(landId, msg.sender, buyer, price);
}

function depositPayment(uint256 landId) external payable
    { TransferRequest storage req = requests[landId];
    require(req.active, "No active transfer");
    require(msg.sender == req.buyer, "Not the buyer");
    require(!req.paid, "Already paid");
    require(msg.value == req.price, "Incorrect amount");
    req.paid = true;
}
```

5.2.6. Approving or Cancelling a Transfer

Once the buyer's payment is in escrow, the sub-registrar verifies the sale documents off-chain and calls approveTransfer. The function confirms that the payment has been made, removes the parcel from the seller's list, assigns it to the buyer, extends the ownership history, and releases the money to the seller, all in a single transaction. If the documents are found to be invalid or the parties withdraw, the owner or a registrar can call cancelTransfer, which unlocks the parcel and refunds any payment held for the buyer. State changes are made before the payment is sent so that the contract is protected against re-entrancy attacks.

```
function approveTransfer(uint256 landId) external onlyRegistrar
    { TransferRequest storage req = requests[landId];
    require(req.active, "No active transfer");
    require(req.paid, "Payment pending");
    Land storage land = lands[landId];
    address seller = land.owner;
    address buyer = req.buyer;
    uint256 amount = req.price;
    _removeFromOwner(seller, landId);
    land.owner = buyer;
    land.underTransfer = false;
    ownerToLands[buyer].push(landId);
    ownershipHistory[landId].push(buyer);
    delete requests[landId];
    (bool ok, ) = payable(seller).call{value: amount}("");
    require(ok, "Payment failed");
    emit LandTransferred(landId, seller, buyer);
}

function cancelTransfer(uint256 landId) external
    { TransferRequest memory req = requests[landId];
    require(req.active, "No active transfer");
    require(
        lands[landId].owner == msg.sender || registrars[msg.sender],
        "Not authorised"
    );
    lands[landId].underTransfer = false;
    delete requests[landId];
    if (req.paid) {
        (bool ok, ) = payable(req.buyer).call{value: req.price}("");
        require(ok, "Refund failed");
    }
    emit TransferCancelled(landId);
}
```

5.3. Deploying and Integrating with a Frontend

After developing and testing the smart contract on Ganache, the next step is to deploy it to an Ethereum test network such as Sepolia before considering any production deployment. The frontend is developed using React.js, with Ethers.js integrated to interact with the blockchain through MetaMask. Registrars can register land and process transfer requests, owners can view their land and initiate a sale, and buyers can verify a parcel and deposit payment through a simple and intuitive interface. Land documents are uploaded to IPFS, and the resulting content hash is stored in the contract. This allows owners, buyers, and officials to use the system without needing technical blockchain knowledge.

5.4. Workflow:

The workflow of the Blockchain-Based Property Ownership and Land Record Management System, as illustrated in Figure 5.1, details the interactions between registrars, land owners, and buyers to ensure the secure registration, transfer, and verification of land.

The process begins with the sub-registrar registering a land parcel in the name of its owner after verifying the documents. The land owner can then view the land records and, when the property is sold, initiate a transfer by naming the buyer and the agreed price. The buyer verifies the ownership of the parcel on the blockchain and deposits the payment into the smart contract. [5]

The sub-registrar then verifies the documents and the payment. As per Figure 5.1, if everything is valid, the ownership of the parcel is transferred to the buyer, the payment is released to the seller, and the land record is updated on the blockchain. If the transfer is found to be invalid, it is rejected and the buyer's payment is refunded.

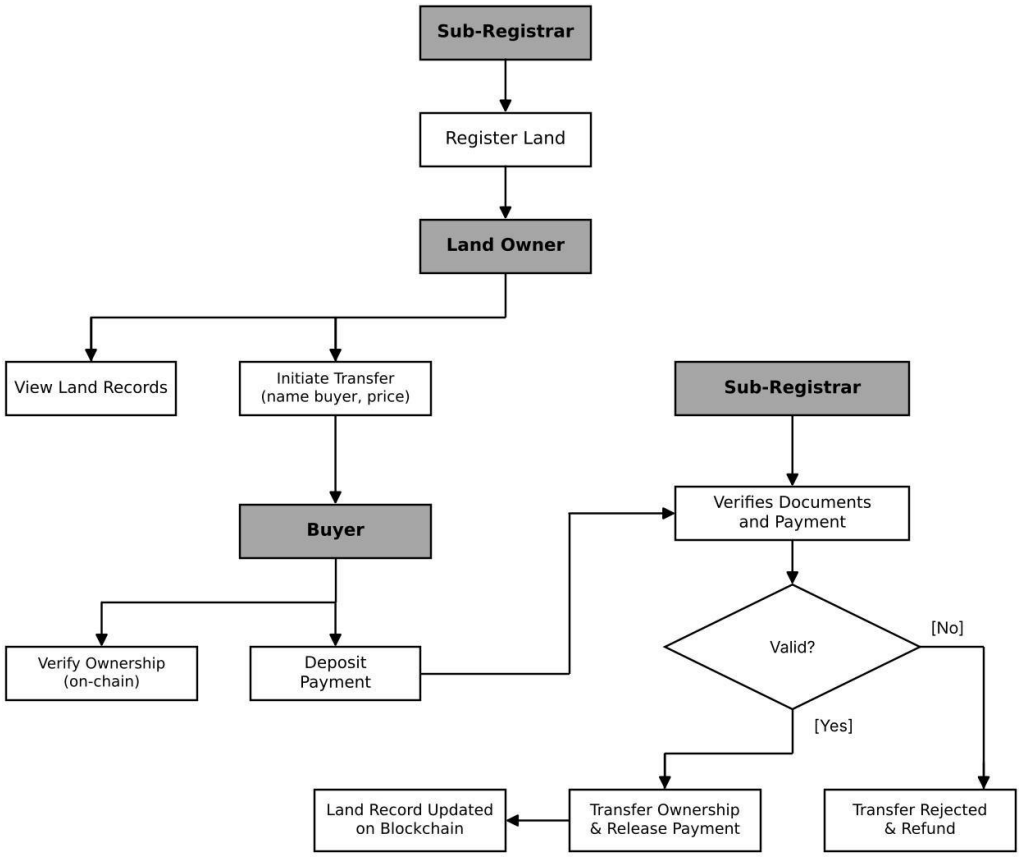


Figure 5.1: Workflow of Blockchain based Land Record Management System

6. BENEFITS

Blockchain technology enhances land record management by introducing transparency, security, and automation. Traditional land record systems rely on paper documents and centralized databases, which are prone to fraud, delay, and loss of data. By leveraging blockchain, land administration becomes tamper-proof, efficient, and easier for citizens to trust. Below are the key advantages:

6.1. Enhanced Security and Immutability

Blockchain provides immutable records, so once a land record is registered and a transfer is completed, it cannot be silently altered or deleted. Every change is a signed transaction that is stored permanently, which protects ownership data against tampering and unauthorized modifications. [3]

6.2. Elimination of Forged and Duplicate Records

Each parcel is stored as a unique record, and each document is represented by its hash on the blockchain. A forged or modified document will not match the stored hash, and a duplicate entry for the same survey number is rejected by the smart contract. This protects buyers and genuine owners from fake claims. [3]

6.3. Prevention of Double Selling

Because the blockchain always shows the current owner of a parcel and locks it while a transfer is in progress, the same property cannot be sold to two buyers at the same time. Once ownership moves to the buyer, any later attempt by the old owner to sell the same parcel is rejected by the contract. [3]

6.4. Transparency and Auditability

All authorized parties can view the same record and the complete history of ownership. Since all transactions are stored on a distributed ledger, any party can verify who owns a parcel, who owned it before, and when each change took place without relying on a single office. This reduces disputes and increases public confidence in the land administration. [4] [7]

6.5. Faster Registration and Mutation

Under the traditional process, the record of rights is updated only after the registration is completed and verified again by another department. In the proposed system, ownership is updated in the same transaction in which the transfer is approved, so the registration record and the ownership record cannot get out of step. This saves time for citizens and officials alike. [7]

6.6. Reduced Corruption and Dependence on Intermediaries

The rules of the transfer are enforced by the smart contract, every action by a registrar is recorded against the registrar’s account, and citizens can check the status of their own records directly. This reduces the scope for unofficial payments and the need to depend on agents to get routine work done. [1]

6.7. Complete Ownership History and Title Traceability

The ownership history of every parcel is stored from the moment it is registered. A buyer or a bank can trace the full chain of title with a single query, which makes the due diligence far simpler and reduces the risk of buying land with a hidden dispute. [7]

6.8. Cost Reduction and Efficiency

Automation of verification and record updating reduces paperwork, storage costs, and the time spent by officials in manual cross-checking. The escrow in the smart contract also removes the need for a trusted third party to hold the buyer's money during the transaction, lowering the cost of each sale.

6.9. Fewer Land Disputes and Greater Citizen Trust

When titles are clear, the history is visible, and records cannot be changed unnoticed, many of the causes of land disputes disappear. Fewer disputes mean fewer cases in the courts and greater trust of citizens in the government's land records, especially among farmers and small landowners who are most affected by disputes. [1]

6.10. Integration and Paperless Land Administration

A blockchain-based land registry can be integrated with other systems such as GIS-based survey maps, digital identity services, banks for loan processing, and tax departments for property tax collection. It also allows departments to move towards paperless administration, reducing the use of paper and the physical storage space needed for record rooms.

Implementing blockchain in land record management brings greater security, efficiency, and transparency to the system. It prevents fraud, reduces administrative effort, and enhances the experience of citizens by making ownership verifiable, transfers automatic, and records permanently available. With the integration of smart contracts and digital identity, it forms a future-proof solution for governments aiming to modernize land administration.

7. CHALLENGES

While blockchain-based land record systems offer numerous advantages, they also come with challenges and limitations that must be considered for effective implementation. Below are some of the key challenges:

7.1. Accuracy of Source Data (Garbage In, Garbage Out)

A blockchain guarantees that a record cannot be changed after it has been stored, but it cannot guarantee that the record was correct when it was entered. If a wrong or fraudulent entry is made while digitizing the old paper records, the blockchain will preserve that error permanently. This concern was raised in reports on the Andhra Pradesh pilot, which noted that records could be tampered with before they are put on the blockchain. [8] Careful verification, survey updates, and a purification of the existing records must therefore be completed before they are migrated.

7.2. Scalability and Transaction Costs

Public blockchains may process only a limited number of transactions per second and can have high gas fees when the network is busy. A state-level land registry handles a very large number of records and transactions, so a public network without scaling solutions may be too slow and costly. Permissioned networks, Layer 2 solutions, or hybrid designs would be needed for real deployment.

7.3. Legal and Regulatory Compliance

In India, the registration of immovable property and the transfer of ownership are governed by laws such as the Registration Act, 1908 and the Transfer of Property Act, 1882, and land is largely a State subject, so procedures vary from state to state. A blockchain record must be given clear legal recognition, and the process must align with the requirements for stamp duty, registration, and mutation. Data protection laws such as the Digital Personal Data Protection Act, 2023 must also be followed when personal details of owners are stored.

7.4. Privacy of Owners' Information

A fully public ledger would expose the names and holdings of landowners to everyone, which can be a safety and privacy risk. Personal details should therefore be kept off-chain or in encrypted form, and a permissioned design with controlled access should be used so that only the necessary information is visible to the public.

7.5. Key Management and User Awareness

Ownership actions are authorized using private keys. If a user loses the key, access to the account can be lost, and if the key is stolen, the thief can act as the owner. Many citizens, especially in rural areas, are not familiar with wallets and digital signatures, so training, simple interfaces in regional languages, and recovery mechanisms such as registrar-assisted account recovery are essential.

7.6. Dependence on Internet Connectivity

Since the system depends on internet access for verification and transactions, users in areas with poor connectivity may face difficulty in accessing and managing their records. Assisted-service kiosks at village and mandal offices can help to reduce this problem.

7.7. Handling Court Orders and Genuine Corrections

Immutability is useful, but there are valid cases such as court orders, inheritance, or correction of clerical errors where a record must be changed. The system must support such changes through authorized, transparent transactions that are added to the history instead of erasing it. Defining who has the authority to make such changes, and how that authority is controlled, is a key design and governance question.

Despite these challenges, blockchain-based land management offers transformative potential in reducing fraud, enhancing transparency, and streamlining land administration. To overcome these limitations, authorities must focus on data verification, legal recognition, privacy protection, user training, and scalable blockchain solutions.

8. CONCLUSION

The report explores the integration of blockchain technology into property ownership and land record management, addressing key challenges in traditional land administration. It highlights the problems of paper-based and centralized records, such as forgery, double selling, corruption, delays in mutation, and the lack of transparency. By leveraging blockchain's immutability and decentralization, the proposed system improves the security of land records and the trust of citizens in them. The use of smart contracts automates the registration, verification, and transfer of land, reducing dependency on intermediaries and delivering a more reliable experience for owners and buyers.

The implementation of the blockchain-based land record system, as described in the report, provides a structured approach to managing land securely. Registrars can register land and approve transfers, owners can view their records and initiate sales, and buyers can verify a parcel and pay safely through escrow. The system ensures that a survey number can be registered only once and that ownership changes only through an approved transaction, preventing misuse. This transition to a blockchain-powered model supports the goals of reducing paperwork, lowering disputes, and promoting transparency in public services.

8.1. Future Outlook for Enhancements

To further improve the system, artificial intelligence (AI) can be used to check scanned documents for signs of forgery and to detect unusual patterns in registrations and transfers. Integration with GIS and satellite imagery can help verify parcel boundaries automatically and detect encroachment. AI-based chatbots in regional languages such as Telugu can guide citizens through verification, transfer, and application procedures.

Integrating decentralized identity (DID) and national digital identity services can ensure that only verified persons can register or transfer land. Links with banks can allow mortgages and loans against land to be recorded and released automatically through smart contracts, building on the involvement of banks in the Swedish land registry pilot. [2] Automatic calculation and collection of stamp duty and property tax within the transfer transaction can further reduce leakage and manual work.

Lastly, scalability solutions such as Layer 2 protocols or hybrid blockchain models that combine a permissioned network with periodic anchoring on a public chain can be adopted to handle large transaction volumes at a low cost while retaining public verifiability. By continuously improving the system with these advancements, blockchain-based land record management can evolve into a secure, efficient, and widely adopted solution for governments.

9. SDG's ADDRESSED

The Blockchain-Based Property Ownership and Land Record Management System aligns with the United Nations Sustainable Development Goal (SDG) 16 by promoting transparency, accountability, and trust in land administration.

SDG 16: Peace, Justice, and Strong Institutions

SDG 16 aims to promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable, and transparent institutions. Secure and reliable land records play an important role in achieving these objectives.

Traditional land record management systems often face challenges such as document forgery, corruption, unauthorized modifications, duplicate registrations, and lengthy verification procedures. These problems can lead to property disputes, financial losses, and a lack of trust in public institutions.

The proposed system uses blockchain technology to maintain secure, transparent, and tamper-resistant property records. Every property transaction is recorded on the blockchain, creating a verifiable history of ownership and property transfers. Once recorded, the information cannot be easily altered without detection, improving the integrity and reliability of land records.

Smart contracts can automate property verification and ownership transfer procedures based on predefined conditions, reducing manual intervention and the risk of fraudulent activities. Authorized users can verify ownership details and transaction histories more efficiently, improving accountability and simplifying record management.

By providing reliable ownership records, enhancing transparency, and reducing opportunities for corruption and fraud, the proposed system helps minimize land disputes and strengthens public confidence in land administration. Therefore, it contributes to **SDG 16: Peace, Justice, and Strong Institutions** by supporting transparent governance, accountable institutions, and a more trustworthy property management system.

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11. APPENDIX A

https://drive.google.com/file/d/1UvxXY3o99kge6UUJ5jLmX2BLqqAuxZsV/view?usp=drive_link